

Chrisil Ouseph

Department of Mathematics, Purdue University, West Lafayette, IN, USA | couseph@purdue.edu

RESEARCH INTERESTS

My research lies at the interface of topology and the structure theory of C^* -algebras, with particular emphasis on homotopy phenomena for Cuntz classes and on the role of regularity properties such as pureness and $\mathcal{Z}$ -stability. I am also interested in topological obstructions and pathological phenomena that arise in the absence of regularity.

EDUCATION

Purdue University

Ph.D. in Mathematics

Advisor: Andrew S. Toms.

Aug. 2022–present

West Lafayette, IN

Indian Institute of Science Education and Research (IISER) Pune

B.S.–M.S. Dual Degree in Mathematics, with Distinction

Thesis: *Algebraic Topology and Obstruction Theory*. Advisor: Vivek M. Mallick.

Aug. 2017–May 2022

Pune, India

RESEARCH PAPERS AND WORK IN PROGRESS

- **Contractible Cuntz classes in $\mathcal{Z}$ -stable C^* -algebras.**
With Andrew S. Toms. Preprint, 2024. arXiv:2412.12366.
- **Homotopies of Constant Cuntz Class in Pure C^* -algebras.**
With Andrew S. Toms. In preparation.

SELECTED TALKS

- **Homotopies of Constant Cuntz Class in Selfless C^* -algebras.** Great Plains Operator Theory Symposium (GPOTS), University of Iowa, Iowa City, IA. May 2026
- **Contractible Cuntz Classes.** Great Plains Operator Theory Symposium (GPOTS), San Juan, Puerto Rico. Oct. 2025
- **Contractible Cuntz Classes.** NSF/CBMS Regional Conference in the Mathematical Sciences: Classifying Amenable Operator Algebras, Texas Christian University, Fort Worth, TX. Jun. 2025
- **The Modern Theory of Classification of C^* -algebras.** Student Operator Algebras Seminar, Purdue University. Apr. 2025
- **Contractible Cuntz Classes.** Operator Algebras Seminar, Purdue University. Feb. 2025
- **Contractible Cuntz Classes.** Interactions between C^* -Algebraic KK-Theory and Homotopy Theory, online. Jan. 2025
- **Types of Operator Convergences and Why Physicists Should Care.** QuTeS Group, University of Murcia, Murcia, Spain. Jun. 2024

RESEARCH ACTIVITIES AND SUPPORT

- Research Assistant, Department Summer Research Grant, Purdue University. Summer 2026
- Research Visit, University of Münster, Münster, Germany. May 2024

SELECTED CONFERENCES AND WORKSHOPS

- Young Mathematicians in C^* -Algebras (YMC*A), University of Oxford, UK (*upcoming*). Aug. 2026
- East Coast Operator Algebras Symposium (ECOAS), North Carolina State University, Raleigh, NC. Nov. 2024
- Oberwolfach Seminar: Classification of C^* -Algebras and Dynamics, Mathematisches Forschungsinstitut Oberwolfach, Germany. May 2024
- East Coast Operator Algebras Symposium (ECOAS), Purdue University, West Lafayette, IN. Oct. 2023
- Groundwork for Operator Algebras Lecture Series (GOALS), Purdue University, West Lafayette, IN. Jun. 2023

TEACHING EXPERIENCE

Purdue University, Indianapolis

- Recitation Instructor, MA 26200: Linear Algebra and Differential Equations. Fall 2025, Spring 2026

Purdue University, West Lafayette

- Instructor, MA 59800: Graduate Preparatory Course for the Real Analysis Qualifying Exam. Summer 2025
- Teaching Assistant, MA 44200: Honors Real Analysis II. Spring 2025
- Instructor, MA 16010: Applied Calculus I. Fall 2024
- Course Coordinator, MA 16200: Plane Analytic Geometry and Calculus II. Summer 2024
- Recitation Instructor, MA 26200: Linear Algebra and Differential Equations. Spring 2024
- Recitation Instructor, MA 26100: Multivariate Calculus. Fall 2023
- Teaching Assistant, MA 30300: Ordinary and Partial Differential Equations. Summer 2023
- Teaching Assistant, MA 36600: Ordinary Differential Equations. Spring 2023
- Recitation Instructor, MA 16100: Plane Analytic Geometry and Calculus I. Fall 2022

SERVICE

- **Co-organizer of AMS Special Session on Recent Advances in Operator Algebras, Chicago, IL** (*upcoming*). Jan. 2027
- **Graduate Mentor, Direct Reading Program, Purdue University.** Supervised Shawanwit Poomsaad on an undergraduate reading project titled *Nets: Convergence in General Topological Spaces*. Fall 2025
- **Co-organizer, Student Operator Algebras Seminar, Purdue University.** Aug. 2024–present